Review notes · for you

Make it automatic: unit check

IoT · Level 3 · Unit check

Materials
Paper and a pen. No device needed.

Learning goals

- IF this, THEN that
- When should it switch?
- A device that remembers
- Power from batteries and the sun

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Quiz sheet

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Name _____ Class _____ Date _____

1 **Which rule runs a classroom fan only when it's needed?**

- ◯ IF above 30 °C OR a class is in, THEN fan on
- ◯ IF above 30 °C AND a class is in, THEN fan on
- ◯ IF above 30 °C, THEN fan on

2 **A smoke alarm should be set so it almost never gives false alarms, even if it sometimes misses a fire.**

◯ True ◯ False

3 **Put the steps for choosing an alarm's threshold in order.**

- ☐ Watch for a few weeks, counting false alarms and misses
- ☐ Log readings on normal days
- ☐ Compare them with readings from times when something went wrong
- ☐ Adjust it, and write down why
- ☐ Put the threshold in the gap between the two

4 **A tank pump is set to switch on below 20% and off above 90%. The tank is at 50% and the pump is running. What happens next?**

- ◯ It clicks on and off around 50%
- ◯ It keeps running until the tank passes 90%
- ◯ It stops, because 50% is above 20%

5 **Match each word to what it means for a sensor's readings.**

Noise	• • • •	One reading that can't be right, like 85 °C at dawn
Glitch	• • • •	The middle reading: one glitch barely moves it
Average	• • • •	The small wobble in every reading
Median	• • • •	Add and divide: calmer, but slower to react

6 **A tank's pump follows one rule: IF the water level is below 20%, THEN pump on. The level sensor breaks and now reads 0% all the time. The pump should keep running, because the tank looks empty.**

◯ True ◯ False

7 **A row in your tank log says only "40". What does it most need?**

- ◯ The time, the unit and which tank
- ◯ A second copy of the number
- ◯ Nothing: the number is enough

8 **A coin cell holds about 220 mAh.**

At an average of 0.1 mA it lasts about _____ hours, which is about _____ days.

Word bank: 92 · 9 · 2200 · 22

9 **Which changes make a Wi-Fi sensor's battery last longer?**

- ☐ Keep the Wi-Fi connected all the time
- ☐ Send only when the reading changes
- ☐ Light a status LED that stays on
- ☐ Send every hour instead of every minute
- ☐ Save ten readings and send them together

10 **A solar-powered sensor needs a battery big enough for the night and for several cloudy days in a row.**

◯ True ◯ False

Check your answers

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1. **IF above 30 °C AND a class is in, THEN fan on**

   AND needs both: the room is hot and someone is there.

2. **False**

   A missed fire can cost lives, so a smoke alarm is set to go off easily, and some false alarms from cooking are the price.

3. **Top to bottom: 4, 1, 2, 5, 3**

   You can't put a threshold in the gap until you know both sides of it, and you can't judge it until it has run for a while.

4. **It keeps running until the tank passes 90%**

   Inside the gap between the two thresholds, the pump keeps doing what it was doing.

5. **Noise — The small wobble in every reading**
   Glitch — One reading that can't be right, like 85 °C at dawn
   Average — Add and divide: calmer, but slower to react
   Median — The middle reading: one glitch barely moves it

   Noise is in every reading, and averaging calms it.

6. **False**

   The rule can't tell a broken sensor from an empty tank, so the pump would run until the tank overflows.

7. **The time, the unit and which tank**

   "40" could be 40%, 40 litres or 40 cm, at any time, in any tank.

8. **At an average of 0.1 mA it lasts about 2200 hours, which is about 92 days.**

   220 ÷ 0.1 = 2,200 hours, and 2,200 ÷ 24 ≈ 92 days, about three months; planned with 70%, about two months.

9. **Send every hour instead of every minute; Save ten readings and send them together; Send only when the reading changes**

   Fewer sends mean less of the radio's big current.

10. **True**

    The sun gives only four to six strong hours a day and can hide for days.